author has studied the effect of the size of the rubber base on the measured contact pressure below the crossed beams.

Errata.—The following correction should be made to the original paper:

Pages 6 and 8: Figures 5 and 6 should be interchanged.

Evaluation of Properties of Rockfill Materials[a]

Discussion by Mosaid Al-Hussaini,[4] A. M. ASCE

The authors have presented valuable information on the deformation and strength characteristics of rockfill material. Probably the most interesting conclusion of the authors is that the angle of internal friction, ϕ', for the rounded Oroville Dam material was about 3° to 4° higher than that of the angular crushed basalt sheared in the triaxial apparatus at confining pressures above 30 psi.

In recent studies (28,29) the Waterways Experiment Station (WES) conducted a number of triaxial compression tests on two granular materials: a highly angular crushed basalt obtained from the Blue Rock Quarry at Napa, Calif. and a rounded sandy gravel material known locally as Painted Rock material, obtained downstream of the saddle dike of Painted Rock Dam at Gila Bend, Ariz. Particles in each material were cleaned, sieved, and recombined to form a straightline gradation curve extending from No. 3 to No. 30 U.S. Standard Sieve. A detailed description of each material has been presented elsewhere (28,29); however, the major physical properties of crushed Napa basalt and Painted Rock material are given in Table 3.

Soil specimens were prepared at two relative densities, 70 % and 100 %, and sheared drained after isotropic consolidation at confining pressures ranging from 60 psi to 500 psi. All specimens were tested in a 2.8-in.-diam triaxial apparatus with enlarged low-friction cap and base. The relationships between angle of internal friction ϕ' and effective confining pressure σ_3' for the crushed Napa basalt and the Painted Rock material are shown in Fig. 13. Fig. 13 indicates that the angle of internal friction for the crushed Napa basalt is higher than comparable Painted Rock material and that the difference decreases with increasing confining pressure. This indicates that angular particles yield higher angles of internal friction than rounded particles under similar testing conditions. Test results by other investigators (31,32,33,34) confirm this conclusion.

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[a] January, 1972, by N. Dean Marachi, Clarence K. Chan, and H. Bolton Seed (Proc. Paper 8672).

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The writer suggests the possibility that testing variables other than the rounded shape of the Oroville Dam material are responsibile for the higher angle of internal friction over that of the crushed basalt.

Table 3.—Physical Properties of Crushed Napa

Physical property	Crushed Napa basalt	Painted Rock material
Specific gravity	2.90	2.66
Maximum void ratio	0.96	0.66
Minimum void ratio	0.53	0.40
Coefficient of uniformity	3.3	3.3
Unconfined compressive	—	—
Strength of intact rock	25,000 psi	—

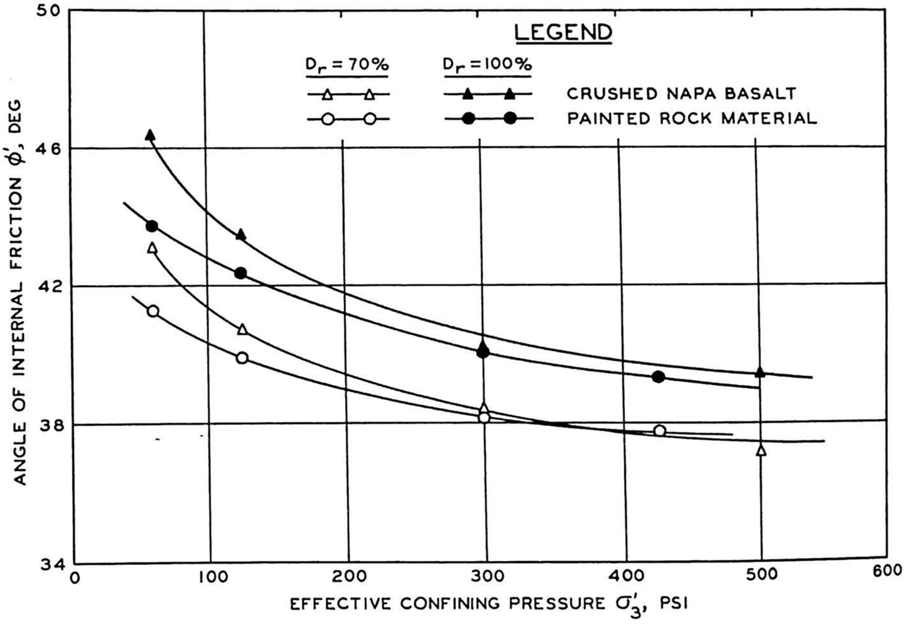

Fig. 13.—Relationships Between Angle of Internal Friction and Effective Confining Pressure for Crushed Napa Basalt and Painted Rock Material

APPENDIX.—REFERENCES

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Discussion by Raymond J. Frost,[5] M. ASCE

The authors describe a problem that continually troubles the soils engineer; namely, what to do with the oversize material that cannot fit into an available testing apparatus. The authors mention two solutions, namely, scalping and parallel grading, but they concentrate only on the latter. Theoretical justification is given for the parallel grading method, on the basis of previous investigations which showed that contact stress and strain between particles is independant of the particle size. While this may be true for quartz, it seems unlikely that the conatact conditions between, e.g., 6-in. particles would be the same as between 0.06-mm particles (No. 200 sieve size). Parallel grading modeling, in effect, increases the amount of the percentage passing No. 200 sieve of the model, and in many instances, this increase in silt and clay fraction can considerably alter the overall shear strength properties. For this reason, the writer does not favor the parallel grading model but prefers one of the several replacement models available.

These replacement models have been described in Ref. 14. The approach consists of replacing the oversize fraction (i.e., the fraction that cannot be accomodated in the test apparatus) by an amount of finer, but still coarse, fraction on the basis that the latter will have either: (1) The same weight; (2) the same cross-sectional area; (3) the same surface area; or (4) the same volume as the coarse oversize fraction omitted.

The transformation equation for (2) and (3) may be derived from geometry as follows:

$$p = \frac{d}{D} \frac{\mathrm{SG}}{\mathrm{sg}} P \qquad (3)$$

in which F = fraction to be replaced; f = replacement fraction; P = percentage of F by weight; p = percentage of f by weight; D = average diameter of F; SG = specific gravity of F; and sg = specific gravity of f.

These concepts are illustrated by the two examples shown in Figs. 14 and 15. Fig. 14 shows various −2-in. models of Oroville Dam material white Fig. 2 shows various −1-in. models of Tarbela Dam Core material. The former shows that scalping and parallel grading models are significantly different while the latter shows that they are similar. It is concluded from Fig. 14 and 15 that more consideration must be given to modeling techniques to arrive at a more universally applicable approach. Of those available, the writer prefers

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